Solution Guide: HR Analytics at Adecco India

Detailed answers and explanations for every case-study question. Companion workbook: `Adecco_HR_Analytics_Solution.xlsx` *(all figures are live formulas on its Data sheet).*

1. Approach

- **Dataset:** 1,470 employees, 35 fields, no missing values. 237 employees left (Attrition = "Yes").
- **Helper columns** added to the Data sheet: `AttritionFlag` (1 if left), `OverTimeFlag`, `SingleFlag` and `JuniorSales` (Sales department and JobLevel 1). These let CORREL and COUNTIFS work on Yes/No fields.
- **Attrition rate by group** = leavers in the group ÷ headcount of the group, computed with COUNTIFS (the formula equivalent of a PivotTable calculated field).
- **"Junior"** is defined as JobLevel 1, the lowest level in the data.
- **Scatter plots:** several hints ask for scatter plots on 1–4 scales or Yes/No data, which render as stripes. The workbook uses banded tables plus CORREL, which show the same relationship more clearly.

2. Basic-Level Solutions

Q1. Overall attrition rate

Method: `=COUNTIF(Attrition,"Yes")/COUNTA(Attrition)*100`

Employees	Left	Attrition rate
1,470	237	**16.12%**

Interpretation: about 1 in 6 employees has left. Compare this with your sector's benchmark, since norms differ by industry; internally it is high enough to be a management concern.

Q2. Department with the highest attrition

Method:
`=COUNTIFS(Department,"Sales",Attrition,"Yes")/COUNTIF(Department,"Sales")`

Department	Headcount	Left	Attrition rate
Sales	446	92	**20.6%**
Human Resources	63	12	19.0%
Research & Development	961	133	13.8%

Interpretation: Sales is the highest. It holds 30% of headcount but 39% of all leavers, which confirms the concern in the scenario. HR is also high but small (63 people). The dataset has three departments, not the engineering/marketing/support split in the narrative.

Q3. Average age of leavers

Method: `=AVERAGEIF(Attrition,"Yes",Age)`

Group	Average age
Left	**33.6**
Stayed	37.6

Interpretation: leavers are about four years younger, so attrition skews to early-career staff. Career-development and early-tenure programmes are the natural response.

Q4. Job satisfaction by job role

Method: `=AVERAGEIF(JobRole,role,JobSatisfaction)` for each role.

Role	Avg satisfaction (1–4)
Human Resources	2.56
Manufacturing Director	2.68
Laboratory Technician	2.69
Research Director	2.70
Manager	2.71
Sales Representative	2.74

Role	Avg satisfaction (1–4)
Sales Executive	2.76
Research Scientist	2.77
Healthcare Representative	2.79
All employees	**2.73**

Interpretation: satisfaction is flat across roles (range 2.56–2.79). HR is lowest. Sales Representatives sit near the average yet have the highest attrition of any role (about 40%), so survey satisfaction alone does not explain who leaves.

Q5. Gender and attrition

Method: COUNTIFS by gender, then a two-proportion z-test (equivalent to a t-test on the 0/1 attrition flag): `z = (p1 - p2) / SQRT(p(1-p)(1/n1 + 1/n2))`, `p-value = 2*(1 - NORMSDIST(ABS(z)))`.

Gender	Headcount	Left	Attrition rate
Female	588	87	14.8%
Male	882	150	17.0%

z = −1.13, p = 0.26.

Interpretation: men leave slightly more often, but the difference is **not statistically significant** at the 5% level. Gender is not an attrition driver here, so retention actions should not be gender-targeted.

Q6. Average monthly income of leavers

Method: `=AVERAGEIF(Attrition,"Yes",MonthlyIncome)`

Group	Avg monthly income
Left	**4,787**
Stayed	6,833

Interpretation: leavers earn about 30% less. Much of this reflects junior job level, but it points to pay and incentives at the entry level as a retention lever. Income is in the dataset's own currency units.

Q7. Distance from home

Method: COUNTIFS on distance bands, plus `=CORREL(DistanceFromHome,AttritionFlag)`.

Distance (km)	Headcount	Left	Attrition rate
1–5	632	87	13.8%
6–10	394	57	14.5%
11–15	115	25	21.7%
16–20	125	23	18.4%
21–29	204	45	22.1%

Correlation = 0.078. Leavers live 10.6 km away on average vs 8.9 km for stayers.

Interpretation: attrition is about 14% up to 10 km and 18–22% beyond. The linear correlation is weak, so distance is a secondary factor. Hybrid days or commute support could help long-distance staff.

Q8. Performance rating distribution

Method: `=COUNTIF(PerformanceRating,rating)` and a column chart.

Rating	Employees	Share
1	0	0%
2	0	0%
3	1,244	84.6%
4	226	15.4%

Interpretation: only ratings 3 and 4 are used. The scale is compressed, so ratings barely separate strong and weak performers or predict who leaves. HR should recalibrate the rating process.

Q9. Overtime

Method: `=COUNTIF(OverTime,"Yes")`

Overtime	Headcount	Left	Attrition rate
Yes	**416** (28.3% of staff)	127	**30.5%**
No	1,054	110	10.4%

Interpretation: more than a quarter of staff work overtime, and they leave at nearly three times the rate of others. Overtime is the strongest single attrition signal in the data.

Q10. Average tenure

Method: `=AVERAGE(YearsAtCompany)`

Group	Avg years at company
All employees	**7.01**
Left	5.13
Stayed	7.37

Interpretation: leavers have about two years less tenure, so risk is concentrated in the early years.

3. Medium-Level Solutions

Q1. Top three factors behind attrition

Method: `=CORREL(variable,AttritionFlag)` for 25 variables, rank by absolute value with `RANK`, then pull the top three with `INDEX/MATCH`.

Rank	Factor	Correlation	Reading
1	OverTime (Yes)	+0.246	Overtime raises attrition
2	MaritalStatus = Single	+0.175	Singles leave more
3	TotalWorkingYears	−0.171	Fewer working years, more attrition

Close behind: JobLevel (−0.169), YearsInCurrentRole (−0.161), MonthlyIncome (−0.160), Age (−0.159).

Interpretation: the pattern is "junior, overworked and early-career". All correlations are modest (below 0.25), so attrition has many causes and no single lever will fix it. Correlation is not causation.

Q2. Job involvement

Involvement	Headcount	Left	Attrition rate
1	83	28	**33.7%**
2	375	71	18.9%
3	868	125	14.4%
4	144	13	**9.0%**

Interpretation: a clear gradient. Staff with the lowest involvement are about four times as likely to leave as the most involved. Involvement is a useful early-warning measure.

Q3. Age and job satisfaction

Age band	Headcount	Avg satisfaction
18–25	123	2.73
26–35	606	2.74
36–45	468	2.73
46–60	273	2.70

CORREL(Age, JobSatisfaction) = −0.005.

Interpretation: no relationship. Satisfaction is the same in every age band, so age-specific satisfaction programmes are not supported. Age matters for attrition through career stage, not through satisfaction.

Q4. Marital status

Status	Headcount	Left	Attrition rate
Single	470	120	**25.5%**
Married	673	84	12.5%
Divorced	327	33	10.1%

Interpretation: singles leave at about twice the rate of married staff. Singles also tend to be younger and more mobile, so this overlaps with career stage. Use it to prioritise support, not to treat individuals differently.

Q5. Training

Sessions last year	Headcount	Attrition rate
0	54	**27.8%**
1	71	12.7%
2	547	17.9%
3	491	14.1%
4	123	21.1%
5	119	11.8%
6	65	**9.2%**

Interpretation: staff with no training leave most (small group, so interpret with care) and those with 5–6 sessions leave least. The pattern is not perfectly linear, but under-trained staff are at higher risk.

Q6. Work-life balance and performance

WLB (1–4)	Headcount	Avg performance rating
1	80	3.18
2	344	3.14
3	893	3.16
4	153	3.15

CORREL = 0.003.

Interpretation: no relationship with rated performance. Work-life balance still matters for retention: staff with WLB 1 leave at about 31% vs 14–18% for the others.

Q7. Stock options

Stock option level	Headcount	Left	Attrition rate
0	631	154	**24.4%**
1	596	56	9.4%
2	158	12	**7.6%**
3	85	15	17.6%

Interpretation: having any option grant at level 1 or 2 cuts attrition by roughly 60–70% compared with none. Level 3 rebounds (small group of 85), so the benefit seems to level off. Extending modest grants to at-risk juniors looks cost-effective.

Q8. Tenure and job satisfaction

Tenure (years)	Headcount	Avg satisfaction
0–2	342	2.76
3–5	434	2.66
6–10	448	2.78
11–20	180	2.76

Tenure (years)	Headcount	Avg satisfaction
21–40	66	2.62

CORREL = −0.004.

Interpretation: no meaningful relationship. Long-serving staff are neither happier nor unhappier than newcomers.

Q9. Business travel

Travel	Headcount	Avg satisfaction	Attrition rate
Non-Travel	150	2.79	8.0%
Travel Rarely	1,043	2.70	15.0%
Travel Frequently	277	2.79	**24.9%**

Interpretation: travel has no visible effect on satisfaction scores, yet frequent travellers leave three times as often as non-travellers. The survey is probably missing travel-related burnout; exit interviews should probe this.

Q10. Years since promotion and performance

Years since promotion	Headcount	Avg performance rating
0	581	3.17
1–2	516	3.13
3–5	158	3.17
6–10	149	3.13
11–15	66	3.23

CORREL = 0.018.

Interpretation: essentially none. Rated performance does not seem to drive promotion timing, which is another sign the rating system is not differentiating.

4. Scenario Deep-Dive: Junior Sales Employees

Attrition by job level within Sales

Level	Headcount	Left	Attrition rate
1	76	32	**42.1%**
2	240	37	15.4%
3	83	17	20.5%
4	34	4	11.8%
5	13	2	15.4%

By role within Sales: Sales Representative 39.8% (33 of 83), Sales Executive 17.5% (57 of 326), Manager 5.4% (2 of 37).

Overtime among Level-1 sales staff: with overtime 72.7% (16 of 22); without 29.6% (16 of 54). The sample is small, but the direction matches the company-wide overtime effect.

Junior sales vs everyone else

Metric	Junior Sales (L1)	All others
Headcount	76	1,394
Attrition rate	42.1%	14.7%
Avg monthly income	2,507	6,721
Avg age	29.4	37.3
Avg job satisfaction	2.67	2.73
No stock options	53.9%	42.3%
Single	46.1%	31.2%
Travel frequently	28.9%	18.3%

Interpretation: the hot spot in the scenario is confirmed. Junior sales staff are young, paid far less, more often single, more often without stock options and more often frequent travellers. Their satisfaction scores are close to everyone else's, so the survey does not

flag the risk.

5. Recommendations

#	Action	Target group	Rationale
1	Cap and monitor overtime	Junior Sales and all staff on overtime	Overtime carries about 3x attrition; track it as an early-warning KPI
2	Entry-level pay and incentive review	Sales Representatives, Level 1	Leavers earn about 30% less; add retention bonuses at 12 and 24 months
3	Extend stock/ESOP grants	Level 1–2 with option level 0	24% attrition with no options vs 8–9% at levels 1–2
4	Career path and mentoring	Single, under 35, Level 1	Clear Rep to Executive path in 18–24 months
5	Minimum training commitment	Staff with 0–1 sessions	Zero-training staff leave at 27.8%
6	Travel policy	Frequent travellers	24.9% vs 8% for non-travellers; cap travel days, add recovery days
7	Flexible commute	Commute over 10 km	18–22% attrition vs about 14% for shorter commutes
8	Fix measurement	HR and management	Recalibrate the 1–4 rating scale; add pulse surveys and exit-interview analysis

Do not target: gender, and age- or tenure-based satisfaction programmes. The data shows no meaningful effect.

6. Limitations and Ethics

- Correlations show association, not cause, and effect sizes are modest.
- Several sub-groups are small (for example 22 junior sales staff on overtime), so treat their rates as directional.

- Employee data is personal: anonymise before sharing, restrict access, and use group-level patterns to prioritise support, never to judge individuals.
- Re-run the analysis regularly as new data arrives, and validate findings with HR, managers and employees before acting.

7. Additional Resources

- Microsoft support documentation on PivotTables, COUNTIFS, AVERAGEIFS and CORREL.
- Kaggle dataset page for the HR Analytics Case Study (Bhanu Pratap Biswas).
- Society for Human Resource Management (SHRM) materials on turnover and retention.